



Politecnico di Torino

Qubit Electronics

Lab 1

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Lab submission instructions

Create a folder with the following name:

`gr<nn>_lab1`

where n corresponds to the group number.

Inside, create a folder for each exercise:

`es01`

`es02`

...

Within each folder, include the spice files necessary for the simulation.

In the root of the lab folder, insert the .pdf with the report. *The report's length must be limited to 10 pages.*

At the end of the lab session, create a .zip archive starting from the root folder and name it

`gr<nn>_lab1.zip`

The created archive must be uploaded by one team member on Portale della Didattica, section Elaborati.

Objectives

The primary goal of this lab is to delve into the fundamental properties and behaviors of quantum circuits and their components, focusing on coherence and relaxation times in superconducting qubits. We aim to explore and model the impacts of the environment and circuit design on qubit performance, specifically looking at factors that influence the relaxation time (T_1) and coherence times (T_2). The lab will consist of several exercises starting with a detailed analysis of a qubit's interaction with its electromagnetic environment, followed by practical circuit simulations and analysis using LTspice. These exercises will help in understanding the Purcell effect, investigating the properties of RLC circuits, and examining the Josephson Junction's dynamic response under different conditions.

1 Theoretical Foundation

Coherence poses the most important challenge for developing a solid-state quantum computer. As the dephasing time T_2^* can never exceed twice the relaxation time T_1 , the relaxation time ultimately sets the limit on qubit coherence. Although T_2^* turned out to be small compared to T_1 in the earliest superconducting qubits, steady progress over the past decade has significantly reduced this gap. Recently, the Transmon, a new type of qubit immune to $1/f$ charge noise, is nearly homogeneously broadened ($T_2^* \approx T_1$). Therefore, understanding relaxation mechanisms is becoming critical to further improvements in both T_1 and T_2^* .

Progress in this direction will be based on the **accurate modeling of contributions** to T_1 and the reliable fabrication of many qubits reaching consistent coherence limits.

One of the main advantages of **superconducting qubits** is their strong interaction with the wires of an electrical circuit, making their integration with fast control and readout possible and allowing for large, controllable couplings between widely separated qubits. The large coupling also implies a strong interaction between the qubits and their electromagnetic environment, which can lead to a short T_1 . However, careful control of the coupling to the environment has been shown to allow the prevention of circuit dissipation. Relaxation times have been studied in various superconducting qubits, created with different fabrication techniques, and measured with many readout schemes.

Typically, values of T_1 vary strongly from sample to sample as they can depend on many factors, including materials, fabrication, and the design of both readout and control circuitry. In some instances, a separation of these components has been achieved. However, typically is not easy to understand the limiting factors, and T_1 often varies strongly even among nominally identical qubit samples.

Because of the simple and well-controlled fabrication of the qubit and the surrounding circuitry, involving only two lithography layers and a single cavity for both the control and the readout, we can reliably understand and predict qubit lifetimes. This understanding extends to a wide variety of different qubit and cavity parameters.

The relaxation times are set by spontaneous emission through the cavity, called the

Purcell effect, or a shared intrinsic limit consistent with a lossy dielectric. Surprisingly, relaxation times are often limited by electromagnetic modes of the circuit, which are far detuned from the qubit frequency. Generally, any discrete-level system coupled to the continuum of electromagnetic field modes is subject to radiative decay. Modifying the states' density can strongly enhance or suppress the emission rate. This effect is named after **Purcell**, who first considered this phenomenon in the context of a two-level system coupled to external circuitry. In atomic physics, the same effect is obtained by placing the atom in a cavity, which affords protection from spontaneous emission.

Considering the case in which all modes of the cavity, except the fundamental one, are far from the transition frequency of the atom, the higher modes can be neglected, thus using an approximation that leads to a Purcell factor of

$$\gamma_k = \frac{g^2}{\Delta^2} \kappa \quad (1)$$

, where g denotes the coupling between qubit and cavity mode, their mutual detuning, and the average photon loss rate. However, this approximation is inadequate for solid-state systems with strong coupling, and treatment going beyond a single mode is necessary.

We can arrive at the same physics using a circuit model, which can subsequently be extended to include higher harmonics. For concreteness, we consider the case of a qubit capacitively coupled to an arbitrary environment with impedance $Z_0(\omega)$.

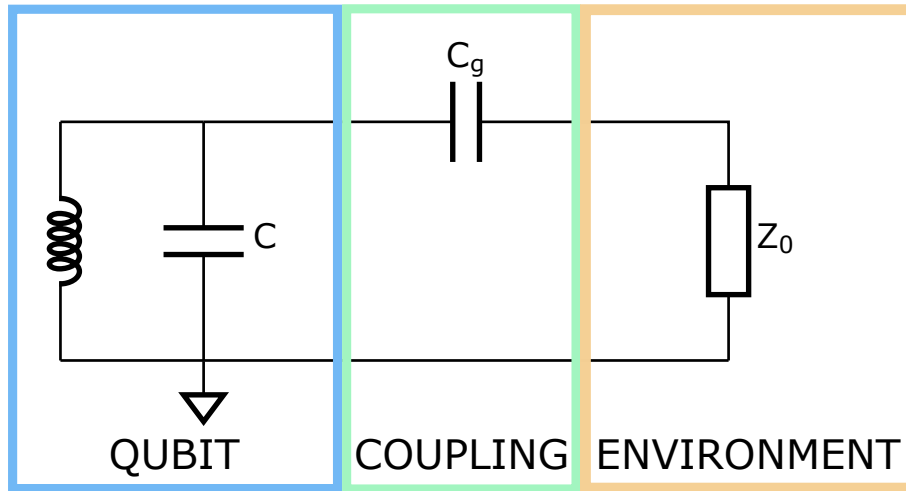


Figure 1: Physic Model

This circuit may be reduced to a qubit coupled to an effective dissipative element, that is, a resistance R . This model will be employed during this laboratory.

Specifically, replacing the coupling capacitor C_g and the environment impedance Z_0 by an effective resistor R , defined as:

$$R = \frac{1}{\text{Re}\{Y(\omega)\}} \quad (2)$$

It is possible to define T_1 as:

$$T_1 = RC \quad (3)$$

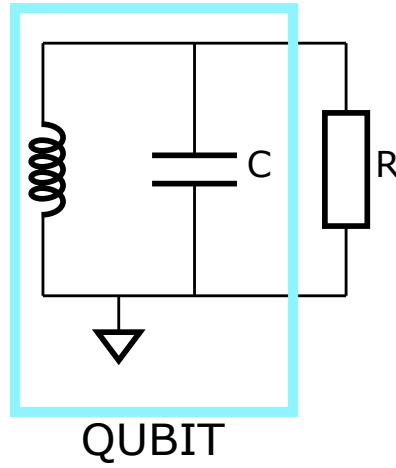


Figure 2: Representation of the qubit with the dissipative element R considered.

where C is the qubit capacitance. Choosing a purely resistive environment $Z_0 = 50\Omega$, yields a decay rate γ :

$$\gamma \approx \frac{\omega^2 Z_0 C_g^2}{C} \quad (4)$$

If, instead, it is coupled to a parallel LRC resonator, the calculated radiation rate can be reduced to

$$\gamma_k = \left(\frac{g}{\Delta}\right)^2 \kappa \quad (5)$$

, thus reproducing the Purcell effect with the single-mode approximation.

2 RLC parallel circuit analysis

This exercise focuses on analyzing an RLC circuit as shown in Figure 3. Draw this circuit in LTspice with the following component values:

- Source current: 1A AC
- Inductance (L): 30nH
- Resistance (R): 100k Ω
- Capacitance (C): 50fF

Firstly, calculate the ideal resonance frequency and the quality factor (Q) associated with this circuit. The equations for resonance frequency (f_R) and Q factor are given by Equations 6 and 7 respectively:

$$f_R = \frac{1}{2\pi\sqrt{LC}} \quad (6)$$

$$Q = R\sqrt{\frac{C}{L}} \quad (7)$$

Then, perform an AC analysis of the circuit using the following LTspice command:

```
.ac dec 10000 1G 20G
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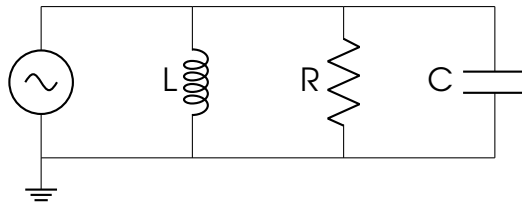


Figure 3: RLC parallel circuit

Measure the ratio between the voltage across the specified node and the current flowing in the current source. How can you describe what you see?

Next, zoom in to identify the peak and determine the frequency associated with it (the resonance frequency) along with its corresponding amplitude value.

Subsequently, find the two frequencies at which the amplitude value is approximately equal to the peak amplitude minus 3dB. Calculate the frequency difference between these two values (Δf).

Finally, calculate Q as $\frac{f_R}{\Delta f}$. Compare this result with your expectations. Does it align with what you anticipated?

3 Analysis of RLC parallel circuit changing the R, L and C values

Considering the previous circuit, measure the Q and f_r values both in Spice and using the corresponding equations of the following circuits.

- $L=30\text{nH}$, $R=200\text{k}\Omega$ and $C=50\text{fF}$
- $L=30\text{nH}$, $R=50\text{k}\Omega$ and $C=50\text{fF}$
- $L=60\text{nH}$, $R=100\text{k}\Omega$ and $C=50\text{fF}$
- $L=15\text{nH}$, $R=100\text{k}\Omega$ and $C=50\text{fF}$
- $L=30\text{nH}$, $R=100\text{k}\Omega$ and $C=100\text{fF}$
- $L=30\text{nH}$, $R=100\text{k}\Omega$ and $C=25\text{fF}$

For each circuit report the Q and f_R values calculated by hand and obtained on Spice.

4 Design an RLC parallel circuit

Considering the previous exercise, design a RLC parallel circuit with the following constraints:

- $C = 50\text{fF}$
- $f_R = 5\text{GHz}$
- $Q = 10^4$

Analyze the voltage on the resistance R and check the frequency resonance and the Q factor.

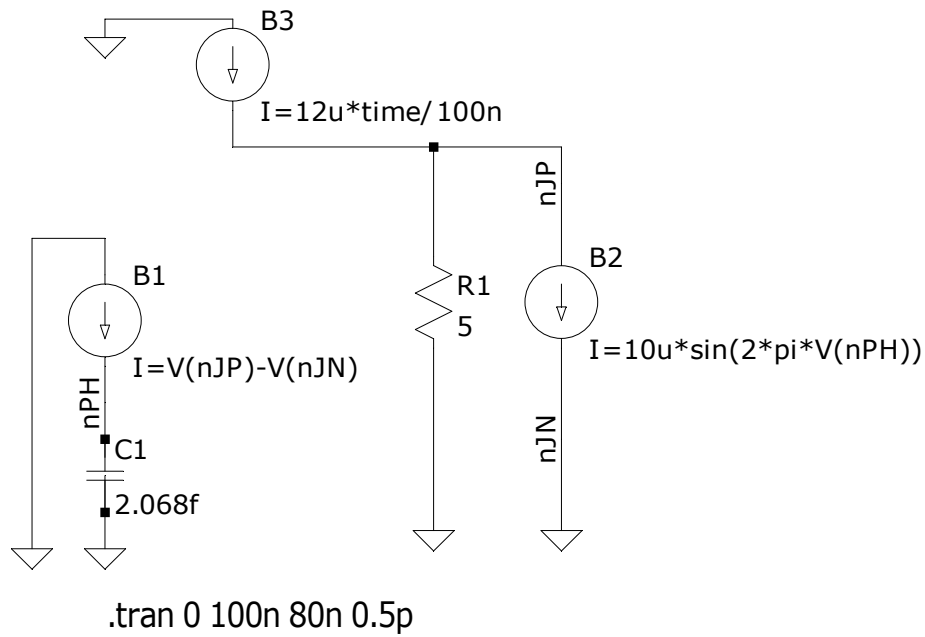
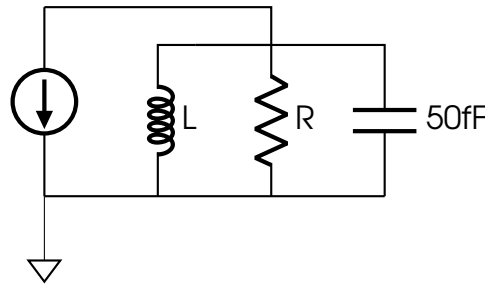


Figure 4: Model of Josephson Junction



- Find the R and L values
- Plot the voltage across the resistance as a function of the frequency
- Check the Q factor and the resonance frequency on the plot

5 Josephson Junction

The objective of this exercise is to analyze a model representing the Josephson Junction. Firstly, we simulate the circuit depicted in Figure 4 using Spice. The circuit consists of a Josephson Junction model.

Next, we observe the voltage at the node nJP and identify the times at which the first 5 peaks occur in the voltage waveform. Using the cursor, we measure the time difference between each consecutive peak and record these values.

We then analyze these time differences. Do they exhibit any noticeable patterns or irregularities? Are they consistent with our expectations based on the behavior of Josephson Junctions?

6 Analysis of the circuit

This exercise is conducted to simulate the behavior of a Josephson Junction. Utilizing the circuit depicted in Figure 5, analyze the behavior by varying the inductance of

the qubit with values of 20.45nH, 23.45nH, and 26.45nH. Are any changes observable between the three circuits behaviors? Determine the frequencies at which peaks occur.

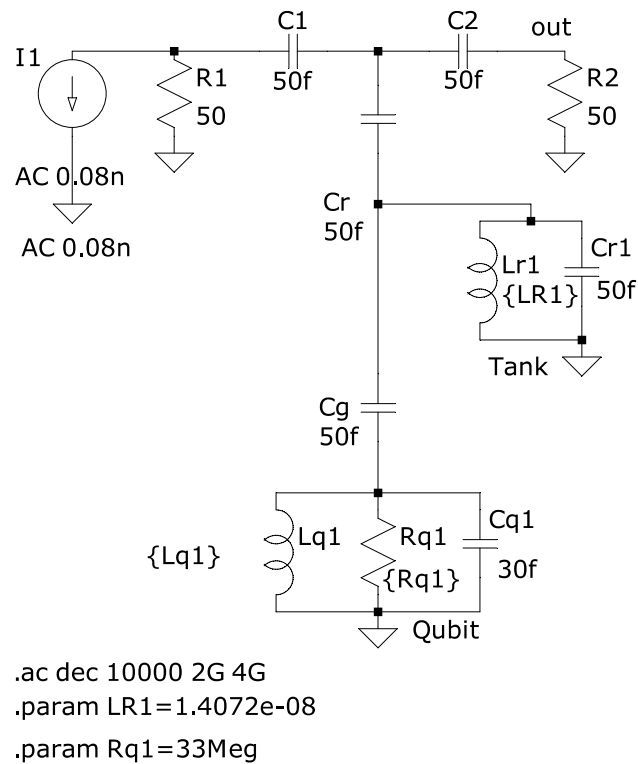


Figure 5: Spice Model of a Qubit

7 Extra: Design of the Qubit characteristic

Define the following circuit using LT-Spice having $\omega_r/2\pi$ that changes from 6 to 11 GHz with 250MHz steps and $\omega_q/2\pi = 6GHz$.

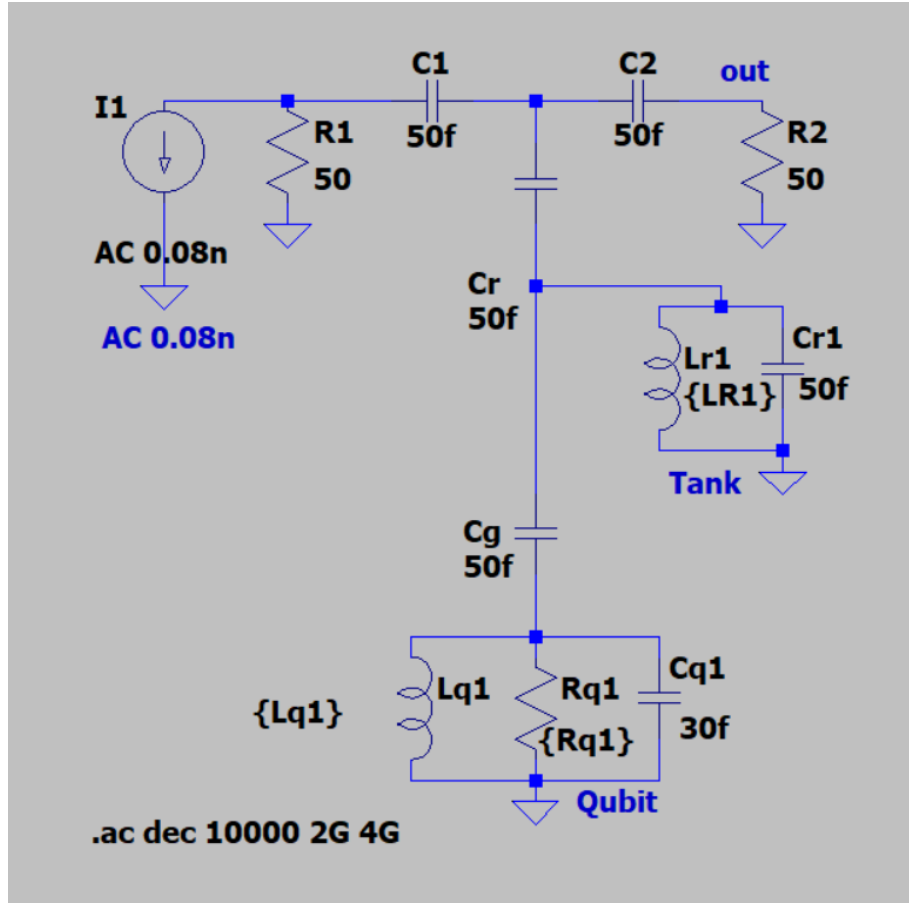


Figure 6: Circuit

- Define the $Lq1$, $Rq1$ and $Lr1$ given the constraints
- Plot the $S_{21} = \frac{V_{out}}{V_{in}}$ as function of the frequency, where V_{in} is the voltage across $R1$ and V_{out} the voltage on $R2$.
- Analyze the behavior and the resonance frequencies obtained.

8 Extra: Uncertainty

Use a Montecarlo simulation considering a variation of the capacitance C_g of the 5% over 100 samples. What are the differences with respect to the previous exercise?